Supporting Information

An agentic framework for X-ray absorption spectroscopy of catalytic interfaces

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Sections S1–S7 preserve the frozen 87-configuration study and its historical evidence. The subsequent 34/98-configuration comparisons retain the earlier paired-study snapshot. Expanded-cohort definitions and results appear in the final sections; the historical scores are not substituted into the current analysis.

S1 Interface construction and structural labels

The lateral interface construction represents two distinct fcc metals with matching (111) facets. Metal identities and row counts define the ordered stripe geometry. Adsorption regions distinguish boundary rows, nearby rows, and terrace regions. Available placement controls include single-binding adsorbates, multidentate species, and coupled carbon-containing intermediates. The broader Cu–Au sampling plan includes 1:1, 2:1, 1:2, 3:1, and 1:3 stripe ratios as well as pure-metal reference surfaces. These ratios describe the construction space; they do not imply that every parent or adsorbate combination is present in the completed spectral cohort.

S1.1 Periodic matching and imposed strain

For an fcc(111) domain with bulk lattice parameter a_i, the natural repeat parallel to the interface is $p_i = a_i/\sqrt{2}$. Integer repetitions n_i give natural lengths $L_i = n_i p_i$. Balanced matching assigns the common length $L = (L_{\mathrm{Cu}} + L_{\mathrm{Au}})/2$, with signed engineering strain

$$\epsilon_i = \frac{L - L_i}{L_i}, \qquad \epsilon_{\max} = \max_i |\epsilon_i|. \tag{S1}$$

The automatic structure-generation route searches $1 \le n_i \le 10$, minimizes $\epsilon_{\max}$, and breaks ties by the total number of repeats. Using the stored values $a_{\mathrm{Cu}} = 3.615$ Å and $a_{\mathrm{Au}} = 4.078$ Å selects $(n_{\mathrm{Cu}}, n_{\mathrm{Au}}) = (9, 8)$ and $L = 23.037185$ Å. The imposed strains are $+0.136776\%$ in Cu and -0.136403% in Au. Balanced one-to-one matching would impose $+6.404\%$ and -5.677%, illustrating the importance of commensurate repeats.

The strain-bounded slab-construction route instead selects the smallest $n_{\mathrm{Cu}} + n_{\mathrm{Au}}$ satisfying a configurable $\epsilon_{\max} \le \epsilon_{\mathrm{allowed}}$, with a default bound of 5% and a search through 12 repeats per domain. It uses strain and repeat asymmetry as subsequent tie breakers. With balanced matching and the same lattice constants, this distinct objective selects $(6, 5)$, with strains -2.997% and $+3.188\%$. A domain-pinned length can be requested in this route. A bound is a construction parameter, not a physical threshold below which strain has no chemical effect. The search objective, lattice constants, common length, repeats, and signed domain strains are therefore part of the parent-model provenance. The transverse stripe-row ratio is an independent construction choice; it is not the interface-parallel repeat ratio.

The surface coordinates used in Figure 2 have an interface-parallel cell length of 23.037185 Å, consistent with the 9:8 balanced repeat. The quoted matching strains describe the imposed initial periodicity; local distortions after relaxation depend on the actual calculation geometry. The top-view rendering retains all 36 Cu and 32 Au atoms in the outermost metal layer, including its height variations; the side view retains all 180 Cu and 160 Au atoms. Adsorbates are omitted only from these substrate illustrations. The views preserve these coordinates up to equivalent in-plane periodic images and use the same shaded atom representation as the other structural panels.

Molecular atom correspondence is retained through relaxation. Chemical labels distinguish the requested intermediate from the molecular components in the relaxed calculation structure. Initial molecule mappings are reconciled with the saved atom order, and local metal layers and contacts are refreshed after relaxation. Geometric screening considers short contacts, periodic images, molecular formulas, bonds and angles, unintended coupling between molecules, and placement relative to the surface. The general interface checker uses periodicity in the surface plane and a nonperiodic vacuum direction. Its distance rules are structural screening criteria rather than electronic bond-order or thermodynamic assignments.

The completed-dataset component analysis uses the C/O/H connectivity in the actual structure used for spectral calculation. Pair cutoffs are 1.85, 1.70, 1.30, 1.20, 0.90, and 1.65 Å for C–C, C–O, C–H, O–H, H–H, and O–O, respectively. Verified molecular identities require compatible bond graphs, not elemental composition alone. An OCCO assignment requires a C–C bond, two C–O bonds, no O–O bond, and the corresponding carbon and oxygen degrees. The position of hydrogen distinguishes CHO from COH and COCHO from OCCOH. Assignments must remain unchanged when cutoffs are scaled by 0.9 and 1.1; ambiguous components are retained for review rather than used as identity targets. The shared audit uses in-plane minimum images and a nonperiodic vacuum direction. The original benchmark predictions remain archived; current-cohort predictions and counts are supplied separately below.

For species s, molecular coverage is N_s/N_{ref}, where N_{ref} is the declared surface reference population. The reference and molecular counts are retained separately from the number of occupied metal sites. For OCCO plus spectator CO, carbon-equivalent coverage is

$$\theta_{\mathrm{C}} = (2N_{\mathrm{OCCO}} + N_{\mathrm{CO}})/N_{\mathrm{ref}}. \tag{S2}$$

Thus two CO molecules and one OCCO molecule can have equal carbon loading but different molecular coverages. Coadsorbed species are represented by simultaneous species-presence labels. COCO is a legacy alias of the bonded OCCO construction template; the unbonded comparator is denoted 2CO. Observed connectivity determines the relaxed-state label.

S2 Electronic-structure and spectral calculations

Structural relaxation is performed with VASP.[10] The committed Cu–Au relaxation examples use a 520 eV plane-wave cutoff, electronic convergence of 10^{-5} eV, a force threshold of 0.05 eV Å^{-1}, fixed cell vectors, and a Γ-centered $1 \times 1 \times 1$ mesh. The adjacent-CO example fixes the bottom three metal layers and relaxes the top two. A dipole correction is applied normal to the asymmetric slab. These are input-example parameters; the completed-cohort records retain the available per-calculation settings.

The present inference dataset uses FDMNES Green-function multiple-scattering calculations.[9] Parsed settings for the representative CO configurations include self-consistency, a 6.0 Å cluster radius, the Perdew option, photon-energy output, and a relative energy range of −5 to 50 eV in 0.2 eV steps. The recorded convolved spectra span 8974.0–9029.0 eV with 276 points. Available record fields do not independently specify every core-hole or broadening parameter. The spectra analyzed here are the saved convolved outputs; no new broadening or energy alignment was fitted during model evaluation.

S2.1 Sensitivity to the number of fixed metal layers

The two relaxation variants share a 340-atom clean Cu–Au parent, the same cell and element order, and identical bottom-two-layer coordinates. Selective-dynamics masks confirm that both metals have three fixed layers in one variant and two fixed layers in the other. Atomwise differences use minimum images in the surface plane and a nonperiodic vacuum direction, without removing a rigid translation. The uppermost interlayer spacing is defined separately for each metal as the difference between its mean heights in layers 5 and 4. Table S1 reports the resulting comparison.

Table S1: Structural sensitivity of the clean parent to fixing two or three bottom layers. Distances are in Å. Δd_{54} and $\Delta \bar{z}_5$ denote the two-fixed minus three-fixed value.

Metal	d_{54}, 3 fixed	d_{54}, 2 fixed	Δd_{54}	$\Delta \bar{z}_5$
Cu	2.1062	2.1109	0.0047	−0.0313
Au	2.4815	2.4739	−0.0076	0.1380

The uppermost interlayer spacings differ by less than 0.008 Å, while the Au domain undergoes a larger mean vertical displacement. Accordingly, the spacing comparison supports a limited structural-insensitivity statement for this parent and does not establish identical local geometries, spectral convergence, or adsorption-energy convergence. Both relaxed structures, the atom correspondence, and the reproducible comparison script accompany the manuscript.

The archived layer inputs contain SCF, Green, and Perdew keywords but no explicit SCFexc, Screening, nohole, or Gamma_hole setting. The output banners identify a 28 July 2026 program revision. The available user guide describes a ground-state self-consistent calculation without a core hole at the K edge, followed by introduction of the hole and screening charge for the absorption calculation.[23] This is an inference from the inputs and documented defaults, not an explicitly recorded core-hole specification; the available guide is dated November 2022. Of the 348 layer spectra, 332 have separate convolution logs reporting Gamma_hole = 1.55 eV and Gamma_max = 10 eV; the other 16 do not have separate convolution logs. These settings therefore cannot be asserted from logs for every layer. A matching guide or source revision is unavailable, so the core-hole prescription is not treated as independently established for that executable. The benchmark reference logs instead identify 11 April 2025 and must not be substituted for the ML-cohort logs.

The framework also generates FEFF and VASP spectroscopy inputs.[8, 11] Shared structural linkage does not imply identical absorber populations or equivalent numerical approximations between backends. The numerical inference scores in the main paper concern FDMNES Cu K-edge data only. Carbon and oxygen edges and other backends form separate potential comparison channels.

S3 Data quality and absorber-population checks

The source inventory contains 172 locations and 155 unique physical structures. Identical source geometries are counted once, including when multiple directories refer to the same structure. Calculation copies within site or layer folders are not counted as new independent configurations. For the full inventory, an unreadable or empty relaxed-structure file can be replaced by a readable initial structure; its presence does not establish a completed relaxation.

A completed spectral record must provide finite, nonconstant Cu K-edge arrays for layers 2–5 on compatible energy grids. Layer ordering is checked against absorber heights, the physical top Cu sites, and the position of adsorbates. Spectral arrays are compared with their saved convolved output, while coordinates, element identities, and atom indices are checked against the relaxed simulation structure. The six FDMNES lattice parameters are reconciled with the relaxed cell before coordinates are compared in a common frame; matching fractional coordinates alone is insufficient. Length and angle tolerances are 2×10^{-5} Å and 2×10^{-5} degrees, respectively. Records with

inconsistent chemical identities or unresolved molecular assignments are not accepted as training labels.

The final convolution population is reconciled with the layer absorber table. This is necessary because a completed directory can include both full-layer calculations and later subset calculations. In two CO configurations, the final convolution represents 36 sites although subset main inputs or intermediate logs describe three sites. Full-population assignment was established by matching the final convolution contributions to their raw spectra, using an energy-cutoff tolerance of approximately 0.00051 eV, complete atom coverage, and compatible grids. These cases are included. A separate CO configuration with an incompatible layer-5 grid is excluded.

Seven records required corrections to molecular identities, counts, or site-species labels. Measurements were unchanged. Duplicate geometries and identical complete spectral stacks were removed before cross-validation. Table S2 summarizes the resulting cohort.

Table S2: Dataset accounting.

Quantity	Count
Source locations	172
Unique inventory structures	155
Completed-record files	102
Duplicate records removed	14
Grid-incompatible record excluded	1
Independent configurations retained	87
Layer spectra retained	348
Mixed-species configurations	9
Single-species configurations	78

S4 Spectral population weighting and feature construction

Let $\mu_l(E)$ denote the raw mean cross section for a layer containing n_l represented Cu sites. Before normalization, the spectrum for a chosen set D of layers is

$$\mu_D(E) = \frac{\sum_{l\in D} n_l \mu_l(E)}{\sum_{l\in D} n_l}. \tag{S3}$$

The seven fixed-depth representations are the four individual layers and averages over the top two, top three, and four available layers. The last representation excludes the unsampled bottom layer and contains no additional attenuation or detection-mode weighting.

For each representation, the minimum intensity is subtracted and the result is normalized by its integrated area. Feature families are: normalized intensity on the recorded energy grid; cumulative trapezoidal integral; first derivative; concatenated smoothed shape, cumulative integral, first derivative, and second derivative; and the first 24 type-II orthonormal discrete-cosine coefficients. Savitzky–Golay smoothing and derivatives use an 11-point window with a third-order polynomial and the energy-grid spacing. The area and transformations are calculated separately for each sample.

Using $\widetilde{\mu}(E)$ for the minimum-subtracted, unit-area-normalized spectrum, the figure labels correspond to

$$\mathrm{Norm}_k = \widetilde{\mu}(E_k), \tag{S4}$$

$$\mathrm{CDF}_k = \int_{E_{\min}}^{E_k} \widetilde{\mu}(E)\,\mathrm{d}E, \tag{S5}$$

$$\mathrm{d1}_k = \left.\frac{\mathrm{d}\widetilde{\mu}}{\mathrm{d}E}\right|_{E_k}. \tag{S6}$$

The numerical derivative uses the stated Savitzky–Golay polynomial fit. Combined concatenates the Savitzky–Golay-smoothed normalized spectrum, CDF, d1, and the second derivative. DCT24 contains coefficients $c_0, \ldots, c_{23}$ of the type-II orthonormal discrete cosine transform of the normalized spectrum; the zero-frequency coefficient is included. The transform follows the type-II cosine basis.[24] The 24 retained coefficients are a fixed representation dimension in the candidate set, rather than an independently optimized physical cutoff. For N energy samples $x_j = \widetilde{\mu}(E_j)$, the coefficients are

$$c_k = \alpha_k \sum_{j=0}^{N-1} x_j \cos\left[\frac{\pi k(2j+1)}{2N}\right], \qquad \alpha_0 = N^{-1/2}, \quad \alpha_{k>0} = \sqrt{2/N}, \quad k = 0, \ldots, 23. \tag{S7}$$

The representations are alternative numerical views of each spectrum, not additional independent samples.

Additional tree candidates use multiscale spectrum/derivative channels with windows of 7, 11, and 21 points, or the surface spectrum together with a surface-minus-subsurface contrast. The latter channels share the sample's surface-area normalization to preserve relative depth amplitudes. No molecular count, site label, geometry descriptor, or filename is used as a spectral predictor.

S5 Models and nested evaluation

The candidate set includes balanced logistic regression with $C = 1$, an RBF SVM with $C = 3$ and scale-dependent kernel width, multi-output random forests and extra-trees, and independent binary extra-trees for each species.[25] Joint forests use 96 trees; the independent ensembles use 64 trees per species. Random forests use a minimum leaf size of two, while extra-trees use a leaf size of one. Both use square-root feature subsampling. Logistic regression and SVM inputs are standardized using statistics fitted on training data only. A sigmoid of the SVM decision function supplies a thresholdable score rather than a calibrated probability. Constant-label training partitions receive constant scores. PCA is not used in this benchmark.

Five outer folds hold out whole configurations with test sizes of 17, 17, 17, 18, and 18. The original assignments of 85 configurations are retained; two recovered configurations are allocated by label balance and fold size without using spectra or prediction results. Each configuration's four layers and combined-layer variants share the same fold. Three inner folds are drawn within each outer-training partition. Candidate inner assignments are ranked by the number of missing-positive training labels and label imbalance, using training labels alone. Fold assignments are supplied with the analysis outputs.

The historical all-depth selector, retained only as an archived reproduction control, compares independent-species tree representations and thresholds within each outer-training set. The global threshold candidates are 0.15, 0.25, 0.35, 0.45, 0.50, 0.55, 0.65, and 0.75. Guarded species-specific adjustments require at least five positive and ten negative training configurations. Adjustments are restricted to a neighborhood of the global threshold and shrunk halfway toward it; they are retained only when the training-fold selection objective improves. Micro F1 is the primary selection objective. The fixed-depth comparison restricts the candidate set to one depth representation while selecting its features and thresholds.

Micro F1 pools true positives, false positives, and false negatives across species. Macro F1 averages the species-level F1 values over all 11 observed species, including failures on rare labels. Exact identity-set accuracy requires the full predicted species set to match the reference. The historical all-depth selector gives micro/macro F1 of 0.779/0.638 and exact-set accuracy of 0.690; it is not the fixed-top-two analysis reported in the updated main figure. Table S3 gives its species-level results.

The fixed extra-trees/first-derivative/surface combination has the highest observed score among the compared fixed entries: micro/macro F1 of 0.798/0.660 and exact-set accuracy of 0.701. This

Table S3: Primary selector by species. Counts are positive configurations and overlap for mixtures.

Species	Positives	Precision	Recall	F1
CHO	5	0.500	0.200	0.286
CO	30	1.000	0.867	0.929
CO2	3	0.667	0.667	0.667
COCHO	1	0.000	0.000	0.000
COH	6	1.000	0.500	0.667
COOH	8	0.714	0.625	0.667
H	22	0.933	0.636	0.757
O	7	1.000	0.714	0.833
OCCOH	4	1.000	0.500	0.667
OCHO	8	0.857	0.750	0.800
OH	5	1.000	0.600	0.750

ranking uses outer-fold results and is exploratory. Repeated analysis of the same simulated cohort does not provide an external confirmation set.

Conditional 95% intervals use 1500 bootstrap resamples of held-out configurations, with the fitted predictions held fixed. Paired depth differences use the same resampled configurations. The intervals exclude model-refitting uncertainty and are not corrected for multiple comparisons. The random seed is 20261020.

S6 Figure construction and structural comparisons

All new figure elements are generated by the accompanying plotting code. The circular workflow uses vector graphics, and the substrate rendering in Figure 2a preserves the metal coordinates from the representative input, with adsorbates omitted from that view. The molecular gallery is shown separately, with one element legend shared by the whole panel. The archived coupling design preserves the supplied starting coordinates without modifying bond distances. The extended Cu–Au site diagram is an ideal triangular fcc(111) surface with explicitly specified metal identities, adsorbate positions, and site contacts. The Cu–Au bridge has one contact on each side of the indicated boundary. CuCuCu, CuCuAu, and CuAuAu hollows are defined by three explicit metal contacts. Surface and adsorbate coordinates share one oblique projection. The updated Figure 2b assigns each of 135 current adsorbate-bearing spectral configurations to one exclusive single-species or coadsorbed identity set. These comprise 118 Cu–Au and 17 Cu(100) cases, including the Cu(100) water/carbonate controls. Clean references and configurations lacking compatible required-layer spectra or resolved geometry lineage do not enter those bars. The 155-structure historical inventory in Section S3 is a distinct snapshot and is not the denominator of the updated figure. These schematic coordinates are supplied separately from the calculated input structures. The molecular gallery uses schematic geometries with explicitly declared bond graphs; these illustrations represent chemical connectivity rather than optimized bond lengths. All molecular drawings share the same coordinate scale and display radius for atomic symbols; these symbol sizes do not represent ionic or van der Waals radii.

The archived reference panels used reference-v2; the updated reference-v3 comparison is given below. The exact fixed-window 0–1 arrays feed both display and similarities; no separate maximum division or earlier edge-jump scores are mixed with them. No graphical display paths are used for metrics. The layer-resolved panel uses recorded CO spectral arrays. The previous species-present/absent descriptive contrast is omitted from the main figure because its reference contains other adsorbates and cannot isolate an adsorbate-induced change. Its historical arrays are retained for provenance only, not as a clean control.

The carbon-coupling figure uses four actual starting models containing two C and two O atoms. Minimum input C–C distances are approximately 2.52 Å for adjacent CO at the Cu-side interface, 2.46 Å for adjacent CO on the Cu terrace, and 1.45 Å for both OCCO models. These distances are not relaxed bond lengths. A matched spectral test requires the same parent, loading, numerical settings, and absorber-population definitions, together with verified final connectivity. No eligible intact-OCCO positives occur in the current inference cohort.

S7 Reproduction and software integration

The archived held-out predictions and reported tables can be replayed without refitting. A separate natural-language-agent run invoked the archived primary selector on a relocated, byte-hashed 87-configuration export and rebuilt 895 fit-cache entries. It reproduced the primary micro/macro F1 of 0.7791/0.6383 and exact identity-set accuracy of 0.6897. Primary and exploratory-winner prediction arrays matched the archive. Across the broader comparison, 140 of 169 held-out array files matched exactly; 29 differed, predominantly in logistic-regression and support-vector variants, with differences also in some random-forest and macro-selected outputs. The rerun used NumPy 2.4.6 and scikit-learn 1.9.0, compared with archived versions 2.3.5 and 1.8.0. These differences are consistent with dependency sensitivity but do not isolate its cause. The archive was not overwritten. All five newly saved outer-fold models reproduced their held-out predictions after reload. This repeats the original cohort and does not constitute independent confirmation.

The agent and graphical interface call a shared audited-training route. Its archived-primary option is distinct from a separate global-threshold integration recipe. A local execution fixture links simulated scheduler events, a saved relaxed geometry, an existing FDMNES record, and held-out model inference. It tests software handoffs; it does not establish production scheduler execution or generation of new spectra.

The full-cohort export contains the relaxed coordinates, spectral records, absorber tables, calculation inputs and outputs, final convolution evidence, frozen folds, and correction provenance needed to repeat the audit and fitting. Byte hashes protect its link to the archived protocol. Figure-only reproduction uses the accompanying plotted data and requires neither model fitting nor access to the simulation tree. Both modes preserve the distinction between the primary nested selector and the exploratory fixed winner.

The training-most-common-set baseline is fitted separately in each outer-training partition and gives micro/macro F1 of 0.323/0.047 and exact-set accuracy of 0.241. The training-majority binary baseline predicts no species and gives zero for all three measures. All baseline results use the frozen configuration folds.

S8 Earlier paired-cohort reconstruction and corrected records

This and the following paired-score sections preserve the earlier 34/98-configuration snapshot. They are historical controls; expanded-cohort membership and scores are defined in the final sections. The new audit uses 116 usable completed configurations (99 Cu–Au and 17 Cu(100)), with two physical top layers each. All 232 native final convolutions are reproduced from saved electronic source spectra and original decks. Source hashes, header multiplicities, physical atom indices, E_{cut}, E_{psii}, shifts, generated decks and numerical residuals are retained per layer. Twenty-eight layers have nonunit source multiplicities, including the clean interface and seven adsorbed Cu–Au configurations. The aggregate source multiplicity is not a physical atom count.

The raw site cross-section is symmetry summed. Where the convolution source list already includes every physical Cu atom, the corrected per-Cu signal is

$$\mu_l(E) = \frac{1}{N_l}\,\mathcal{C}\left[\sum_{j\in l}\frac{x_j(E)}{m_j}\right], \tag{S8}$$

where m_j is the recorded source multiplicity and $\mathcal{C}$ retains the original source-specific energy convention and final convolution prescription. Explicit unit-weight controls reproduce the native implicit source sum. Inverse-multiplicity weights recover one contribution per physical absorber. This is an audited source convention, not inference from atom-count uniqueness alone. Layers without nonunit multiplicities retain their original scale.

The uploaded reconstruction tool is an October 22, 2026 official FDMNES package; its hashes and source URL are pinned. The saved electronic outputs identify July 28, 2026. Numerical reproduction of the convolution does not prove that later source documents the earlier electronic core-hole default. Core-hole metadata therefore remain `not_reported`; the historical 332/348 separate broadening logs are not silently extended to the 16 missing logs. No new electronic spectrum or HPC job is generated.

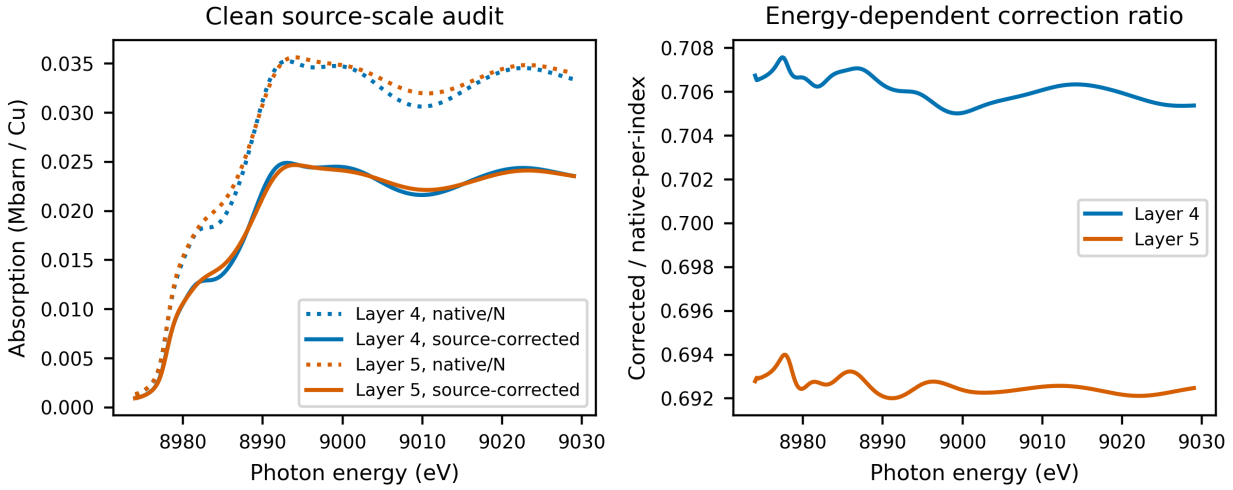


Figure S1: Native aggregate divided by unique atom count versus source-corrected physical per-Cu spectra for the actual clean reference. The ratio varies with energy, precluding correction by a guessed scalar. Dotted lines denote native-per-index; solid lines denote source-resolved correction.

S9 Pair definitions, signed features and units

The primary 34 cases have identical initial metal coordinates, cell, composition and constraints to the clean model; this is strong model compatibility, without independently proven historic ancestry. Sixty-four weaker same-substrate/model cases are kept separate. The pair manifest records paths, SHA-256, energy support, physical populations, settings, observed labels, matching evidence and exclusion status. One clean-model family serves all eligible pairs; 66 distinct metal-coordinate arrangements are not asserted to be independent parent families. No matched OCHO support is inferred from a same-cell reference.

The common supported grid is 8974–9029 eV in 0.2 eV steps, with linear interpolation and no extrapolation. Physical Cu–Au populations are 36 per layer in layers 4+5; Cu(100) uses four per layer in 3+4. The top-two combination uses population weights in both members. No Cu(100) clean reference is substituted from Cu–Au, bulk Cu, or an adsorbate-bearing state. Raw electronic and final convolution sources are retained separately. The clean edge-step windows are 8974–8977 and 9018–9025 eV, using one shared divisor for every pair member. No fitted pairwise alignment is applied.

Let $d_j = \Delta\mu(E_j)/S$ and $h = 0.2$ eV. The signed-grid representation contains d_j. A fixed Savitzky–Golay smoother has 11 points and polynomial degree 3; its first and second derivatives use h explicitly, giving units $S^{-1}\mu$/eV and $S^{-1}\mu$/eV^2. The cumulative difference is

$$C_j = \sum_{k=1}^{j} \frac{d_{k-1} + d_k}{2}(E_k - E_{k-1}), \qquad C_0 = 0. \tag{S9}$$

It has relative-intensity eV units and may decrease. It is not a probability CDF. Combined concatenates smoothed signed intensity, cumulative difference and both derivatives. Training-only feature scaling does not rescale a query's amplitude independently.

For N grid points, orthonormal type-II DCT coefficients are

$$c_k = \alpha_k \sum_{j=0}^{N-1} d_j \cos\left[\frac{\pi}{N}\left(j + \frac{1}{2}\right)k\right], \quad \alpha_0 = N^{-1/2}, \quad \alpha_{k>0} = \sqrt{2/N}. \tag{S10}$$

DCT8/16/24 retain $k = 0, \ldots, K-1$, including DC.[24] No per-sample amplitude normalization follows the transform. Unit tests independently check analytic cumulative integrals, linear amplitude response, self-subtraction zero, both signs, DC, incompatible-reference rejection and shared-gain calibration.

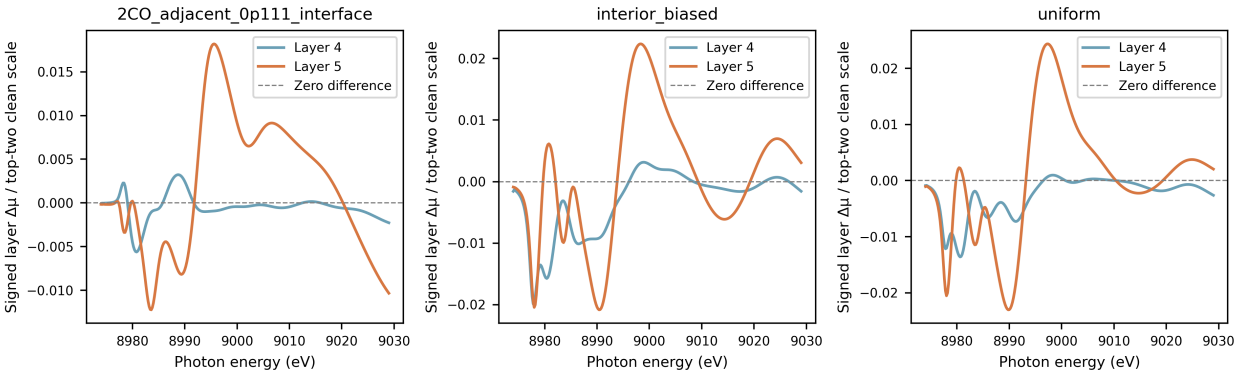


Figure S2: Layer-resolved actual paired differences for the three equal-loading examples in the main text. The same top-two clean scale is used for both layers; their individual amplitudes are not renormalized. Dashed gray lines denote zero.

S10 Earlier paired-cohort grouped evaluation and scores

Outer GroupKFold(5) partitions configurations; inner GroupKFold(3) selects Combined/DCT8/16/24 and compact regularization.
Regression candidates are Ridge $\alpha = 1, 10, 100$ and two leaf-size-2 tree controls. Identity uses independent balanced logistic classifiers with $C = 0.1, 1$ and two compact tree controls. Single-class outer or inner training labels use constant predictions. Inner predictions set per-label thresholds; no outer test label selects features, regularization or thresholds. Nearly constant columns are removed using training data before feature scaling. This prevents constant clean-reference padding from changing random feature selection. All layer/absorber views remain with the configuration. No structural targets enter predictors.

The five representations are compared on identical matched cohorts, and the 98-case weaker reference set is a separate sensitivity study. The fixed one-reference affine control gives equivalent predictions for raw, difference and clean-plus-difference inputs. Per-species supports, scores, every selected recipe and inner prediction array accompany the saved outer predictions and models. The six primary supported labels are CO, H, O, COH, COOH and OH; macro scores over all eleven targets are not interpreted as support for absent chemistry.

Table S4: Identity on identical configurations within each declared cohort.

Scope	Input	Micro F1	Macro F1 (11)	Exact set
matched	legacy norm.	0.7385	0.3259	0.6471
matched	corrected norm.	0.7463	0.3375	0.6471
matched	per-Cu raw	0.7941	0.3635	0.6765
matched	delta	0.7941	0.3635	0.6765
matched	paired	0.7941	0.3635	0.6765
matched	training_label_majority	0.0000	0.0000	0.0000
weaker	legacy norm.	0.6759	0.5584	0.5306
weaker	corrected norm.	0.7109	0.5622	0.6020
weaker	per-Cu raw	0.7426	0.6072	0.6224
weaker	delta	0.7426	0.6072	0.6224
weaker	paired	0.7426	0.6072	0.6224
weaker	training_label_majority	0.0000	0.0000	0.0000

Table S5: CO-only loading relative to all surface metal sites; primary seven cases, weaker-reference sensitivity 25 cases. Missing or negative predictive value is retained.

Scope	Input	MAE	RMSE	R^2
matched	legacy norm.	0.0069	0.0098	0.6715
matched	corrected norm.	0.0076	0.0103	0.6348
matched	per-Cu raw	0.0065	0.0091	0.7143
matched	delta	0.0065	0.0091	0.7143
matched	paired	0.0065	0.0091	0.7143
matched	train mean	0.0169	0.0212	-0.5396
weaker	legacy norm.	0.0144	0.0232	0.9318
weaker	corrected norm.	0.0146	0.0240	0.9270
weaker	per-Cu raw	0.0140	0.0237	0.9288
weaker	delta	0.0140	0.0237	0.9288
weaker	paired	0.0140	0.0237	0.9288
weaker	train mean	0.0658	0.0930	-0.0967

An explicit mixture diagnostic applies the mean of the five saved CO-only fold models to four matched CO-containing mixture configurations, absent from every model fit and selection. Physical raw and differential MAE is 0.02021 CO/all surface metal sites, with $R^2 = -10.31$. Thus the CO-only predictor does not establish mixture coverage inference; coadsorbates change the spectral response. This frozen-model diagnostic is distinct from mixture-trained cross-validation.

Input	Matched mixtures (4) MAE	Weaker mixtures (12) MAE
Legacy normalized	0.01878	0.02221
Corrected normalized	0.01972	0.02326
Physical raw	0.02021	0.02546
Signed difference	0.02021	0.02546
Paired	0.02021	0.02546

Conditional paired intervals resample configurations 1000 times with seed 20261027. They condition on fixed predictions and do not measure refitting, parent transfer or a prospective test. Repeated use of the existing folds is exploratory. A new fit is not independent confirmation. Configuration-level geometry targets are evaluated separately using fixed DCT16/Ridge(10); primary contact targets are constant, and broader hollow compositions lack $CuAu_2/Au_3$ support. Molecular support cannot inflate the configuration sample count.

S11 Earlier Cu(100) augmentation controls

The raw loading control fixes DCT16/StandardScaler/Ridge(10) and the original grouped Cu–Au folds. Four Cu(100) CO configurations use leave-one-configuration-out augmentation with the other three, while a Cu–Au-only control sees none. Targets are CO/surface Cu loading. It reproduces the four previously reported MAEs without changing target denominators. The separate raw identity control uses CO/H/O/OH only, fixed DCT16/balanced logistic $C = 1$ and inner grouped thresholds. Twelve Cu(100) configurations have these supported labels; five H_2O/CO_3 cases are explicit unsupported chemistry, and carbonate connectivity does not establish charge. Domain-balanced summaries must use domain-level scores rather than counting the larger domain as independent transfer evidence.

The earlier export lacked a compatible clean Cu(100) spectrum, so its augmentation tests used raw XAS. The expanded study now reconstructs the clean layers from supplied electronic spectra and evaluates differential inputs; those results are reported in the final sections. Neither conditional configuration augmentation nor one clean family per domain establishes new-parent transfer.

S12 Synthetic robustness protocol and physical interpretation

All stress settings are fixed before examining outer results. Shared rigid energy errors are $\pm 0.2, 0.5, 1, 2$ eV; residual reference-only errors are $\pm 0.2, 0.5, 1$ eV. Shared recalibrated gains, relative adsorbate gain and edge-step errors are $\pm 2, 5\%$. Extra Gaussian $\sigma = 0.25, 0.5, 1$ eV uses a positive normalized finite-domain kernel; native lifetime broadening remains. Independent Gaussian noise amplitudes are 1, 2 and 5% of peak intensity, with three seeds. Shared-reference scenarios reuse the same reference noise realization across query configurations, retaining dependency. This simplified pointwise noise is not an instrument model.

Both pair members are perturbed before subtraction. The fixed interior grid 8976–9027 eV supports every shift without extrapolation. Raw and difference controls use identical cropped grids and outer folds. The compact fixed stress model is DCT16/Ridge(10) for coverage or independent balanced logistic $C = 1$ for identity; thresholds use inner grouped folds. Training-only augmentation uses four deterministic views with shared shifts $-0.5, -0.2, 0.2, 0.5$ eV, relative shift ± 0.2 eV, 1% gain/noise and Gaussian $\sigma = 0.25$ eV. Copies never cross folds; no stress curve selects nuisance settings. All 832 metric rows and corresponding held-out arrays are retained.

The stable-background dilution test uses $a_{\mathrm{obs}} = b + f(a - b)$ with $f = 0.1, 0.25, 0.5, 1$. It reports error against full surface loading and against the response proxy $f\theta$ separately. Neither determines the unknown illuminated surface fraction. Simulated top-two slab spectra do not establish experimental depth weighting. For identity, normalized signed experimental differences require a compatible relative normalization and energy convention, together with assessment of detection geometry, incident-

intensity and self-absorption corrections, noise and residual reference coverage. Absolute per-Cu calibration and an independently known surface fraction are not identity prerequisites. Calibrated absolute coverage requires additional constraints on the measured active absorber population and intensity mapping. No paired labeled experimental accuracy is reported.[27, 28]

S13 Reference benchmark: processing order and complete similarities

The preserved original notebook uses Gaussian $\sigma = 1$ eV and exploratory $+120/+125$ eV shifts; one cell reuses a previous intensity variable, so its empty/unverified inputs cannot silently define a new curve. The new executed notebook records its actual arrays and corrected portable route. Raw $\rightarrow$ native output $\rightarrow$ fixed additional Gaussian (VASP only) $\rightarrow$ rigid alignment $\rightarrow$ common-grid min–max normalization are saved independently. Original experimental negative noise and native raw values remain unchanged; unsupported local pre-edge subtraction is removed before display normalization. Min–max scaling is relative intensity, not a cure for a faulty baseline.

Cu-calibrated offsets are fitted separately for every declared method, including PBE and the GW-labelled dataset. The shape diagnostic uses the first resolved Cu/Cu_2O peak and the early CuO inflection, not an arbitrary later peak. First-feature matching has no fitted width; Gaussian $\sigma = 1$ eV is the notebook prescription rather than measured resolution. The feature search windows, nominal offsets, detected features, support checks and failure status are saved. No label-based outer-model result determines these windows. Chemical-shift accuracy is not inferred from per-compound shape alignment. Cu_2O FEFF retains common Cu calibration because its first feature is unresolved.

The Cu PBE small early peak is aligned at the author-specified 8982 eV anchor. Its saved broadened native maximum lies at 8868.32345 eV, giving a rigid offset of 113.67655 eV. The earlier offset of 124.72175 eV had assigned this feature to the later 8993-eV peak. The correction is -11.04520 eV and leaves the supplied Gaussian width unchanged. The same corrected Cu offset is transferred to Cu_2O PBE in the transferable-offset analysis; the independently aligned Cu_2O comparison is unchanged. All affected intensities and metrics are recalculated from complete saved native-stage arrays, with no extrapolation or fit to similarity scores. Incoming arrays, corrected arrays and the reproducible processing script are retained separately.

All reference scores and plotted arrays use 176 points over 8980–9015 eV. Each trace, including experiment, uses identical-window extrema in $(I - I_{\min})/(I_{\max} - I_{\min})$. Flat/invalid traces and inadequate support are unavailable, not zero similarity. Cosine is $x \cdot y/(\|x\|\|y\|)$ without centering. Pearson correlates centered intensities; Spearman correlates ranks with ties. Official SciPy functions supply the correlations; no pointwise p-values imply independent energy samples. A high correlation cannot validate absolute energy, chemical shifts or a mechanism.

Table S6: Complete available reference similarities. Shift sign is $E_{photon} = E_{native} + shift$. CuO PBE has no valid numeric table and remains missing in both views.

Material	Method	View	Shift/eV	Cosine	r	ρ
Cu	Green	Cu-cal.	0.0452	0.9972	0.9852	0.9329
Cu	FDM	Cu-cal.	3.0452	0.9908	0.9828	0.9313
Cu	FEFF	Cu-cal.	-7.7208	0.9874	0.9665	0.9619
Cu	PBE	Cu-cal.	113.6766	0.8956	0.6039	0.7927
Cu	GW-PAW	Cu-cal.	125.0649	0.9779	0.8777	0.9593
Cu	Green	feature	0.0452	0.9972	0.9852	0.9329
Cu	FDM	feature	3.0452	0.9908	0.9828	0.9313
Cu	FEFF	feature	-7.7208	0.9874	0.9665	0.9619
Cu	PBE	feature	113.6766	0.8956	0.6039	0.7927
Cu	GW-PAW	feature	125.0649	0.9779	0.8777	0.9593
Cu_2O	Green	Cu-cal.	0.0452	0.9953	0.9790	0.9355

Material	Method	View	Shift/eV	Cosine	r	ρ
Cu_2O	FDM	Cu-cal.	3.0452	0.9807	0.8645	0.7164
Cu_2O	FEFF	Cu-cal.	-7.7208	0.9869	0.9099	0.7331
Cu_2O	PBE	Cu-cal.	113.6766	0.8411	0.3254	0.3556
Cu_2O	GW-PAW	Cu-cal.	125.0649	0.8960	0.6469	0.7382
Cu_2O	Green	feature	0.2919	0.9963	0.9776	0.9390
Cu_2O	FDM	feature	-0.7081	0.9978	0.9825	0.9711
Cu_2O	FEFF	feature	-7.7208	0.9869	0.9099	0.7331
Cu_2O	PBE	feature	115.9511	0.9032	0.6446	0.7491
Cu_2O	GW-PAW	feature	125.5837	0.9050	0.6627	0.7443
CuO	Green	Cu-cal.	0.0452	0.9967	0.9754	0.8925
CuO	FDM	Cu-cal.	3.0452	0.9468	0.8010	0.8326
CuO	FEFF	Cu-cal.	-7.7208	0.9872	0.9026	0.7954
CuO	GW-PAW	Cu-cal.	125.0649	0.9602	0.7940	0.7994
CuO	Green	feature	0.8926	0.9963	0.9811	0.9107
CuO	FDM	feature	-0.5074	0.9607	0.7718	0.6316
CuO	FEFF	feature	-6.8164	0.9918	0.9437	0.8597
CuO	GW-PAW	feature	125.2560	0.9614	0.8073	0.8225

The newly supplied Cu_2O OUTCAR and HDF5 arrays independently agree for 2000 energy points and diagonal tensor components. The isotropic signal is their diagonal average of the imaginary output component. Inputs specify PBE+U, $U_{eff} = 6$ eV on Cu, ICORELEVEL=2, CLZ=1 and CH_SIGMA=0.1 eV. Complete numeric table, electronic EDIFF convergence and normal termination footer are recorded separately; a complete table alone is not a convergence certificate. The CuO PBE HDF5 cannot be parsed (bad object-header version) and its supplied OUTCAR lacks a complete numeric table. Original files are uploaded with this limitation. PAW dataset labels and hashes are retained without licensed potential contents. GW-labelled PAW data do not establish a quasiparticle or electron–hole stage.

Timing provenance distinguishes measured CPU from allocated core-time estimates. CuO FDMNES fdm is retained as the author's corrected 129600 CPU s. The separate 91018.9 s log explicitly identifies "Multiple scattering calculation (Green)" and supplies the Green-mode timer. Oxide folder names alone cannot identify the solver; the archived oxide directories are reversed relative to their mode evidence. FEFF module values are CPU times. Cu_2O PBE SCF/XAS CPU times are 2400.813/7149.952 s; GW-labelled stages are 21585.344/8539.898 s. Resources, system sizes and settings differ, and no measured speedup is claimed. The complete timing CSV preserves stage and evidence distinctions.

S14 Earlier figure cohorts and clean-reference comparisons

The following 115/98/34-case tables preserve the earlier snapshot. The current Figure 2b and Figure 4b counts are superseded by the expanded-cohort tables below; the diagnostic species trace plots use the expanded 33 matched single-species cases and broader Cu–Au-only sensitivity set. The updated figures use declared cohorts rather than pooling differently validated configurations. The current convolution reconstruction covers 116 completed configurations: 99 Cu–Au (including one clean reference) and 17 Cu(100). Removing the clean reference leaves 115 adsorbate-bearing configurations for Figure 2b. A geometry mismatch remains excluded. No unfinished checkpoint is counted as a completed spectrum.

The earlier Figure 4b snapshot used the 98-case physically scaled top-two analysis, not a new refit of the 87-case depth benchmark. For class k, its support is $\sum_i y_{ik}$ across pure and mixed configurations. This explains the difference between exclusive bars in Figure 2b and positive supports in Figure 4b: CHO has two single-species configurations plus five CHO/CO/H mixtures. The seven positive configurations provide seven spectra, not seven molecules or 28 independent layer samples.

The current Figure 4e and the following diagnostic traces use 33 initial-model-and-constraint-matched single-species configurations. The six supports are CO 8, COH 3, COOH 3, H 11, O 6 and

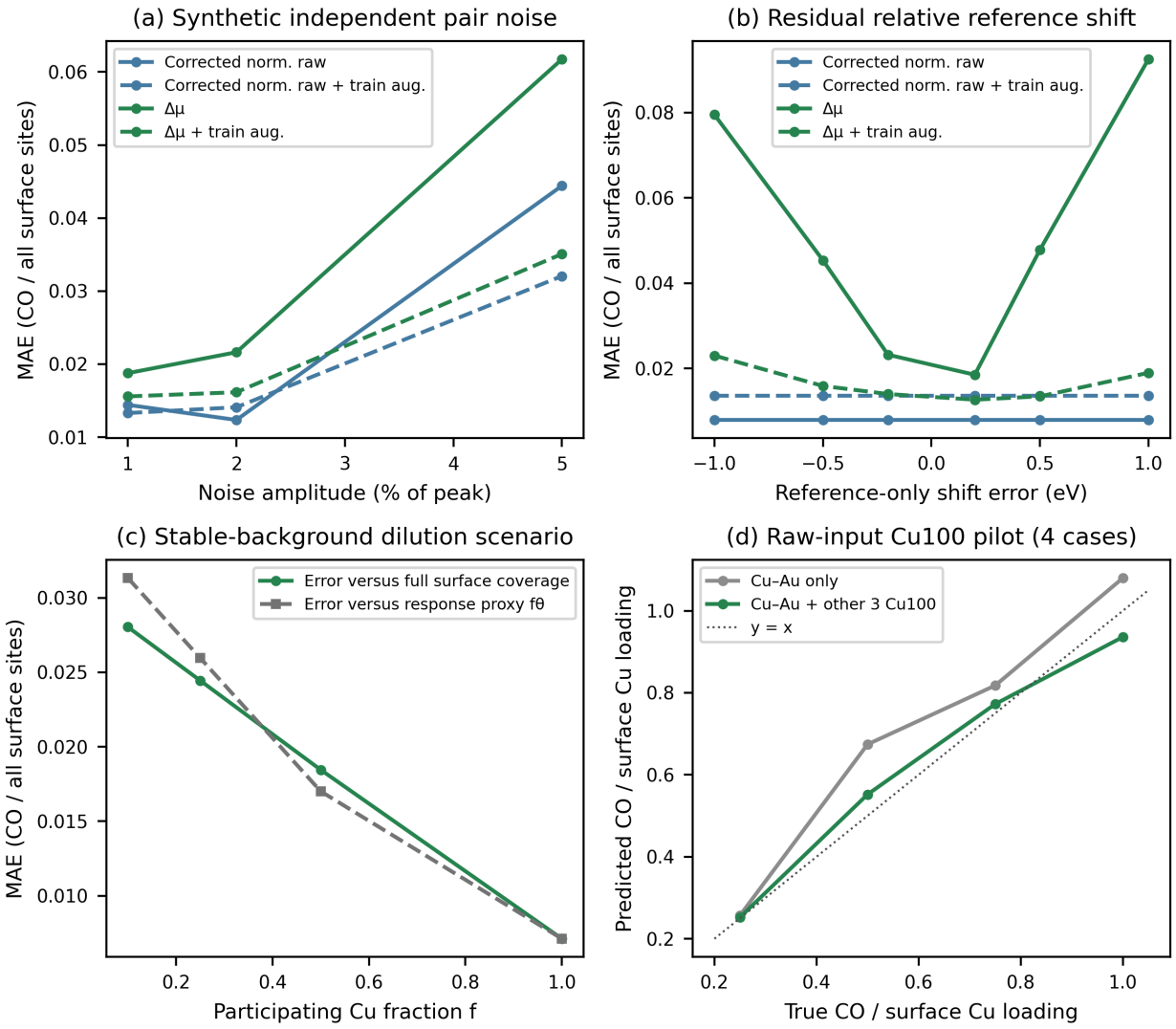


Figure S3: Synthetic robustness scenarios and the separate raw-spectrum Cu(100) loading control. The noise and relative-shift panels compare unaugmented models with training-only augmentation. The dilution panel distinguishes error against full surface loading from error against the response proxy $f\theta$. These simulated perturbations are not measured experimental uncertainties. The raw Cu(100) control belongs to the earlier export; the main text now reports the expanded differential study.

Table S7: Cohort definitions in the updated figures. "Mixed" counts configurations containing more than one observed species.

Cohort	Total	Single	Mixed	Use
Original Cu–Au benchmark	87	78	9	Fig. 4a,c,d
Current Cu–Au, adsorbed	98	86	12	Figs. 2b,4b
Matched Cu–Au subset	34	30	4	Figs. 2b,5a
Matched single-species subset	30	30	0	Fig. 4e
Cu(100), adsorbed	17	13	4	Fig. 2b
Cu(100), four supported labels	12	12	0	Fig. 5d
Cu–Au, CO-only	25	25	0	Fig. 5b
Matched Cu–Au, CO-only	7	7	0	Primary paired loading test
Cu(100), CO-only	4	4	0	Fig. 5c

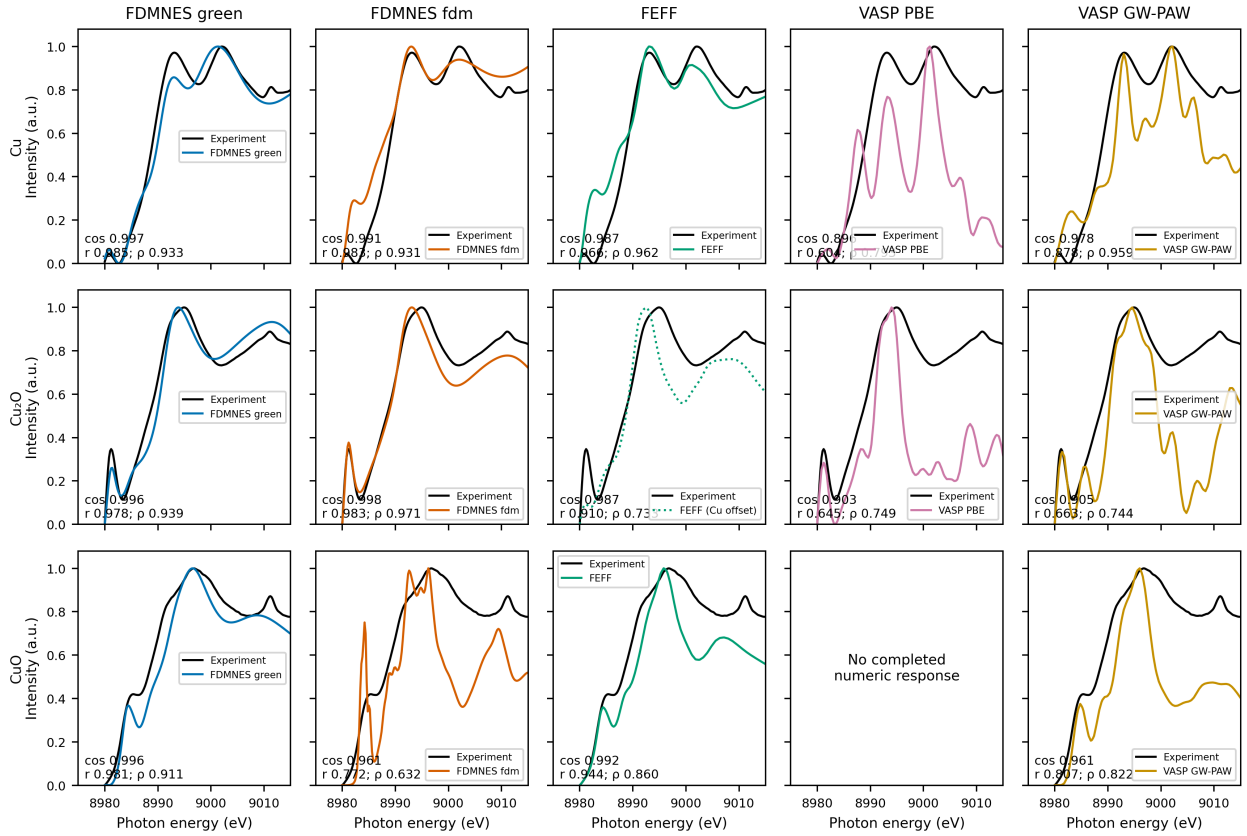


Figure S4: Three-material by five-method first-feature comparison, using actual numeric arrays and the same fixed min–max interval. Black denotes experiment. Missing CuO PBE is blank. The Cu_2O FEFF dotted trace is explicitly identified as retaining transferable Cu calibration.

Table S8: Earlier 98-case top-two raw-spectrum adsorbate identification. Supports count independent positive configurations, including mixtures.

Species	Support	Precision	Recall	F1
CHO	7	0.333	0.143	0.200
CO	37	0.967	0.784	0.866
CO_2	3	0.500	0.667	0.571
COCHO	1	0.000	0.000	0.000
COH	7	1.000	0.857	0.923
COOH	9	0.500	0.444	0.471
H	25	0.944	0.680	0.791
O	7	1.000	0.714	0.833
OCCOH	5	0.667	0.400	0.500
OCHO	8	0.800	0.500	0.615
OH	6	1.000	0.833	0.909

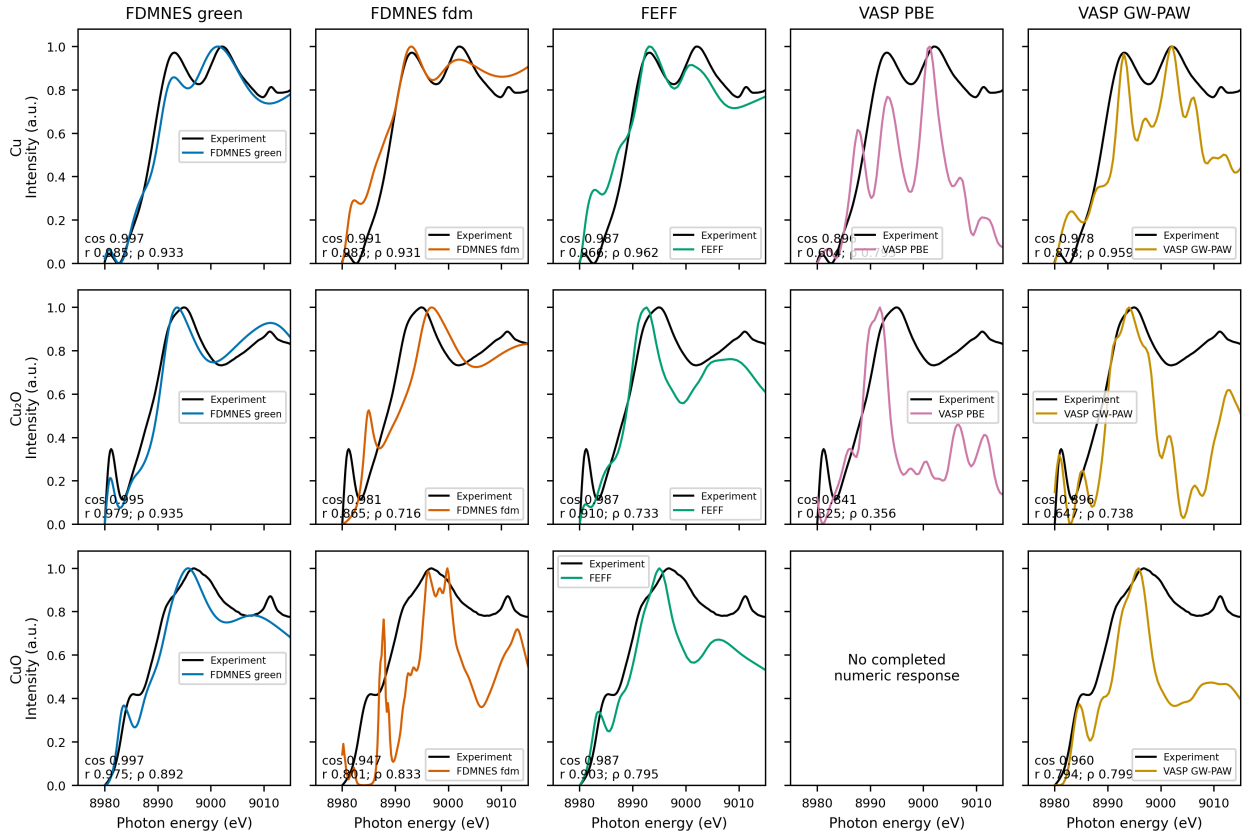


Figure S5: Transferable Cu-calibrated comparison, separately from first-feature shape alignment. Offsets are shared across compounds within each declared method, not forced to be shared between PBE and the GW-labelled dataset.

OH 2. Each class mean is computed from the signed difference in the signed-difference definition in the main text using the same clean-derived scale; the plotted quantity is $100 d_i(E)$. The heatmap uses one fixed ±6% color range without per-class scaling. These means are conditional on the available coverages, geometries and one clean reference family. Between-configuration variation is shown by individual traces below rather than treating the 276 energies as independent observations. The separate same-substrate panel includes all available single-species classes, but does not upgrade their reference matching to that of the primary subset.

Table S9: Latest available logged CPU seconds, with the XAS-stage values used in Figure 3b. Values retain their native accounting scope. For VASP, SCF and XAS are separate stages. A blank indicates no separately available timer. FEFF values sum recorded module CPU timers. FDMNES values are reported total CPU counters, with the author-confirmed CuO finite-difference value used as supplied. These are not harmonized total core-seconds or controlled speedups.

Reference	Method	Preceding SCF / s	XAS route / s	Numeric spectrum
Cu	FDMNES green	—	7,165.200	yes
Cu	FDMNES fdm	—	16,349.200	yes
Cu	FEFF	—	1,368.617	yes
Cu	VASP PBE	139.563	95.148	yes
Cu	VASP GW-PAW	150.584	847.926	yes
Cu_2O	FDMNES green	—	14,086.300	yes
Cu_2O	FDMNES fdm	—	117,943.800	yes
Cu_2O	FEFF	—	1,202.306	yes
Cu_2O	VASP PBE	2,400.813	7,149.952	yes
Cu_2O	VASP GW-PAW	21,585.344	8,539.898	yes
CuO	FDMNES green	—	91,018.900	yes
CuO	FDMNES fdm	—	129,600.000	yes
CuO	FEFF	—	2,872.231	yes
CuO	VASP PBE	5,241.338	—	no
CuO	VASP GW-PAW	17,093.088	4,614.561	yes

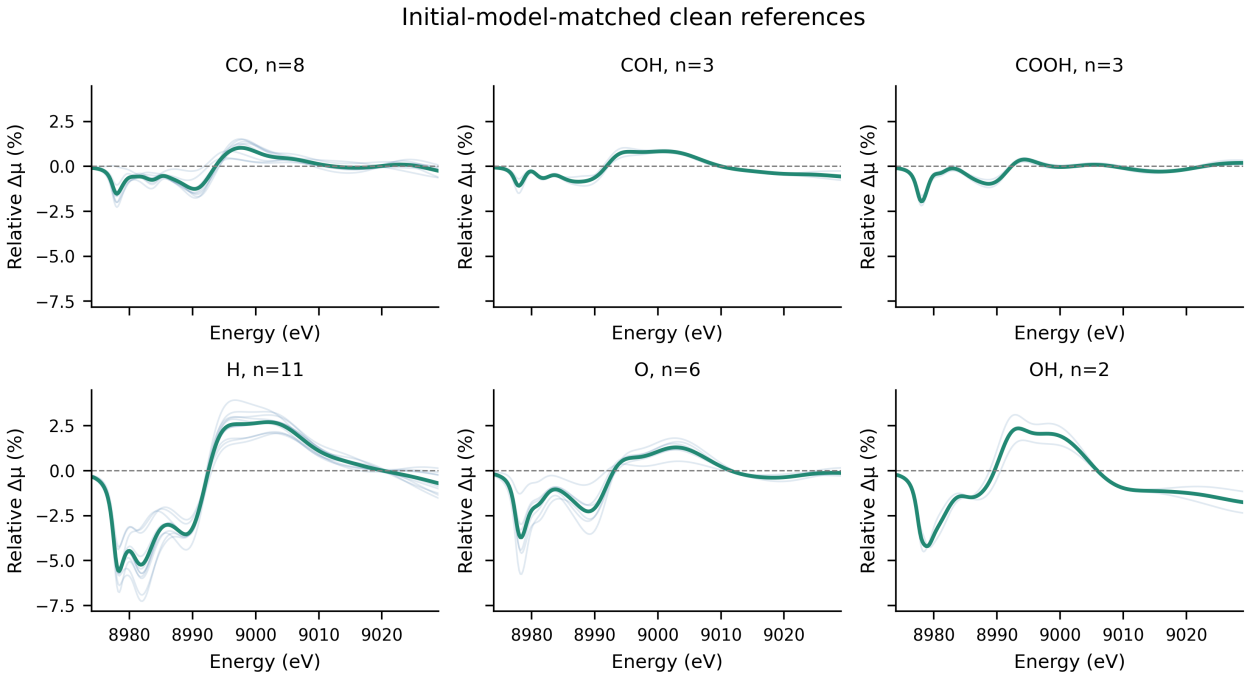


Figure S6: Primary matched-model clean-reference differences. Faint blue curves are individual configurations, and green is the class mean. All curves use the same clean-derived scale. Dashed gray is zero. These are adsorbate-bearing configurations minus the actual separately relaxed clean interface, not species-present/absent or equal-species-mean differences.

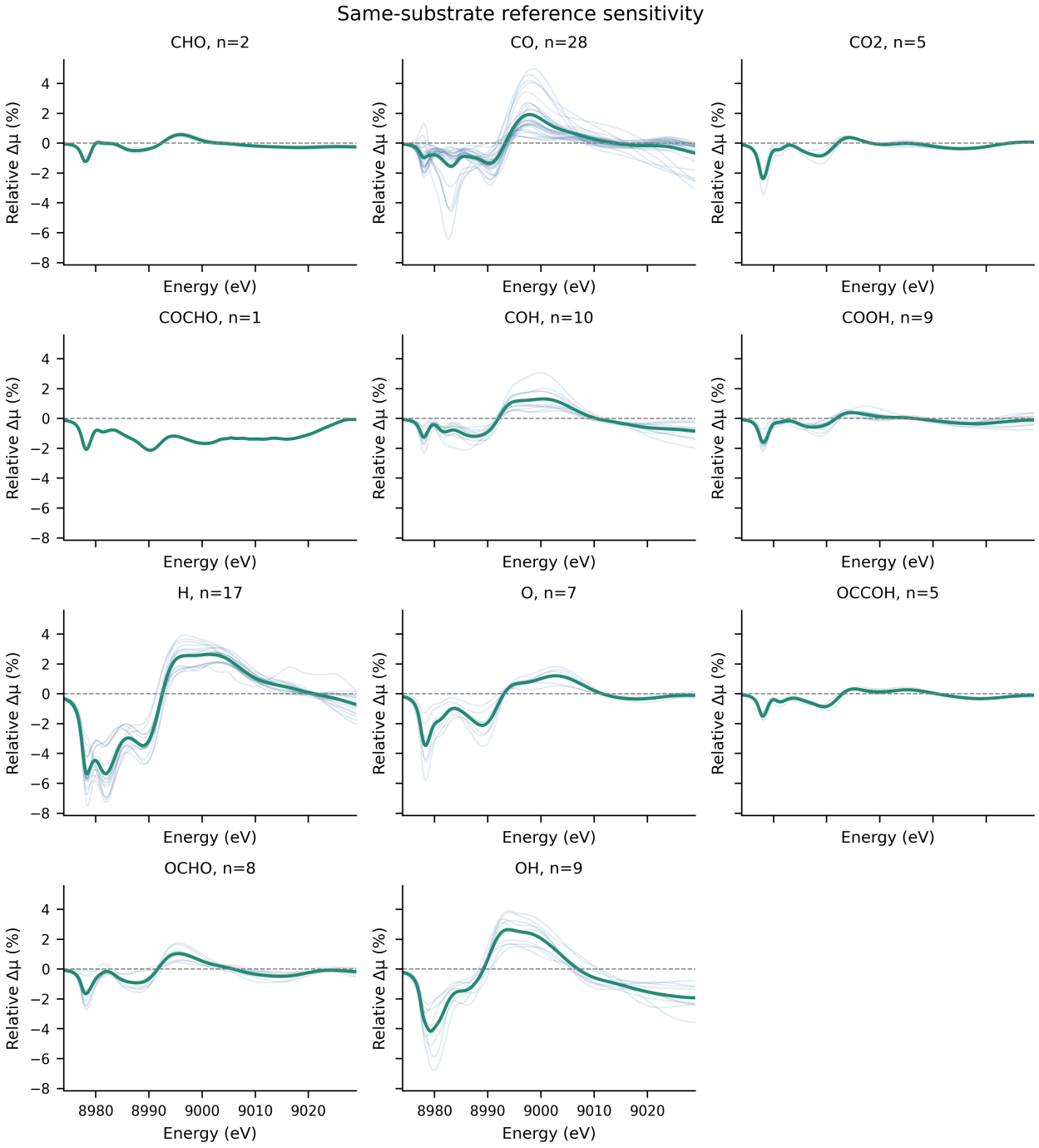


Figure S7: Broader same-substrate clean-reference sensitivity for all single-species classes. The panel combines primary matched and weaker-reference configurations and is separated from the primary result. Line conventions match the preceding figure.

S15 Historical unfinished OCCO–water checkpoints

Read-only snapshots were captured at 2026-10-27 01:45 UTC with source modification times and SHA-256. The actual current CONTCAR is used for identity; restart POSCAR need not be the original seed. Original dimer atoms are tracked element-wise with in-plane minimum images and nonperiodic vacuum. Connectivity is tested over C–C cutoffs 1.65–2.2 Å, with explicit C–O and hydrogen graphs. Water hydrogen bonds are not covalent proton transfer. Two-C versus C,O binding is audited independently of C–C retention. The three cases captured on that date are all dissociated 2CO, not retained C2O2. Carbon flags are free in these checkpoints, rather than artificially fixed dimer bonds. Available restart-local trajectories do not establish a full history before the supplied restart.

Archived case	Step	C–C/Å	C–O/Å	Free force/eV Å^{-1}
2H_2O	16	3.283	1.159–1.165	0.288
2H_2O interior	1	3.105	1.166–1.167	0.280
2H_2O atop–bridge	8	3.476	1.163–1.182	0.291

At capture all three were unfinished, without an ionic convergence message or covalent proton transfer to the original dimer. Last evaluated OUTCAR forces may correspond to the geometry before an optimizer-updated CONTCAR. Fifteen other seed-only alternatives are separately inventoried; their existence is not a running or completed relaxation. Current snapshots, available complete trajectories, OSZICAR and input/output evidence are retained. No snapshot is labeled a stable minimum or a completed training record.

S16 Earlier shared inference and saved evidence

A real local natural-language request calls the shared audited `differential_v1` recipe: validate geometry-linked pairs, fit declared grouped recipes, save/reload, and infer held-out identity, loading and supported configuration-level geometry. One hundred main and forty geometry fold models are reloaded through the same feature API. Code/data hashes, split groups, missing-positive labels and preprocessing/reference conventions are stored. Scores are not asserted to be calibrated probabilities; unsupported chemistry/domain/reference queries are flagged or rejected. No production scheduler or newly generated spectrum is demonstrated.

Archived primary/fixed-winner reproduction remains separate from this current route. The primary and fixed winner match the recorded fresh arrays, 140/169 broader OOF files match, 29 differ, and five recorded fold-model reloads match. These historical results do not prove causation from dependency changes. Proposed missing controls and future configurations have not already improved inference.

S17 Expanded spectral cohort and source correspondence

The expanded record export contains 146 versioned records: 145 converted or merged supplied measurements and one reconstructed clean Cu(100) reference. The source audit verifies numerical correspondence for 553 retained native series and element-aware correspondence between each retained layer geometry and its FDMNES input. The record export is larger than each model cohort because incomplete required layers, uncertain lineage and unsupported molecular identities are treated separately. Record labels follow molecular connectivity in the XAS-used structure, including the distinction between H–C and H–O bonding. A root relaxed structure is not substituted for a different archived geometry that generated an existing spectrum.

Of 137 source-scaled configurations, 135 are adsorbate bearing and two are clean references. Figure 2b includes 118 adsorbed Cu–Au and 17 adsorbed Cu(100) configurations. Closed-label fitting instead uses 117 adsorbed Cu–Au configurations plus the existing clean Cu–Au negative, and 12 supported-label Cu(100) configurations. Thus the two Cu–Au counts of 118 have different membership. One Cu–Au CO/water case and five Cu(100) water/carbonate cases remain audited interference queries. The reconstructed Cu(100) reference is an input, not a thirteenth adsorbate example.

The OH configuration with reliable layers 4+5 is retained only for those layers; incompatible lower layers are removed from its versioned measurement. A roughened OH configuration remains excluded because its required physical-layer partition is ambiguous. The H root/XAS-copy discrepancy retains only the previously established exact archived lineage. The analogous unresolved CO/H discrepancy is excluded. Complete numeric output alone is insufficient when its geometry lineage cannot be established. These exclusions are recorded in the accompanying audit and required-layer tables.

The clean Cu(100) reconstruction uses three nonoverlapping symmetry representatives for four atoms per layer. A cell-metric-preserving x/y reflection maps the multiplicity-two representative to its partner. That multiplicity is retained, unlike full explicit lists that already enumerate every physical site. The symmetry check rejects arbitrary x/y swaps in rectangular cells. Convolution of saved electronic spectra is a local reconstruction, not a new electronic calculation or proof of ionic convergence.

S18 Relative differential intensity and experimental inputs

For a reaction/reference pair, $d = (\mu_a - \mu_r)/S_r$ retains the signed relative response. If both spectra and the reference scale are multiplied by the same unknown factor c, then $(c\mu_a - c\mu_r)/(cS_r) = d$. Absolute cross-sections per Cu atom are consequently not required for adsorbate identity inference. Either a consistently processed pair or a precomputed normalized signed difference may be used, provided its convention is compatible with the model. Energy registration, sign, background treatment, reference state and noise remain relevant. This statement does not imply experimental accuracy for the current simulation-trained models.

The simulation grid is 8974–9029 eV at 0.2 eV increments. Its reference scale is the difference between clean-spectrum averages over 9018–9025 and 8974–8977 eV. This short-window near-edge proxy must not be interpreted as a demonstrated full experimental post-edge normalization. Established work on near-edge normalization motivates checking background and energy-range sensitivity.[30, 31] No independent peak alignment or per-difference minimum, area or norm scaling is applied. Source-weighted physical spectra establish comparable simulated absorber averages before this relative scaling. Absolute loading additionally depends on the mapping between the simulated surface ensemble and the measured active fraction. Identity, relative change and calibrated absolute coverage therefore have distinct input requirements.

For 37 adsorbed Cu–Au cases the initial metal model and constraints match the clean model. The six single-species groups contain CO 8, COH 3, COOH 3, H 11, O 6 and OH 2 configurations,

totaling 33. Four matched mixtures are not used in Figure 4e. Eighty further adsorbed cases have weaker reference correspondence; the clean negative is self-referenced. These compatibility classes are saved by configuration. Only one reference family exists per substrate domain, so new-parent generalization is unmeasured.

S19 Current grouped identity and loading evaluation

All spectra from a configuration remain in the same validation group. The compact comparison uses five outer folds and three grouped inner folds, selecting among Combined or DCT16 features with regularized independent logistic classifiers ($C = 0.1, 1$), or the specified tree classifier. Feature scaling and thresholds are training-only. This six-candidate single-seed study is separate from the three-seed fixed-Combined training-size control below. No model is selected by outer-test performance. Missing-positive training labels and domain-specific supports accompany the predictions.

The native-normalized representation starts from native convolution weights and applies the established minimum-subtracted unit-area transformation. Corrected raw and differential representations use reconstructed source weights and a shared clean-derived intensity scale. The comparison therefore varies both weighting and normalization. A corrected-and-normalized control is required to isolate them. Unit-area shape differences between native and corrected arrays are detectable in 8 of 118 Cu–Au and 6 of 12 Cu(100) configurations, with maximum integrated absolute differences of approximately 0.00185 and 0.00790. These diagnostics do not establish that the small shape changes caused the score differences.

Table S10: Current compact identity comparison. Macro F1 uses all eleven declared labels. The normalized-raw control differs in weighting as well as normalization.

Representation	N	Micro F1	Macro F1	Exact set
Cu–Au / native normalized	118	0.7717	0.5845	0.6186
Cu–Au / corrected raw	118	0.7194	0.5504	0.5763
Cu–Au / $\Delta\mu$	118	0.7194	0.5504	0.5763
Joint / native normalized	130	0.7123	0.5604	0.5769
Joint / corrected raw	130	0.7170	0.5552	0.5692
Joint / $\Delta\mu$	130	0.7361	0.5714	0.6000

Table S11: Positive configuration support and Cu–Au-only differential identity performance. Mixtures contribute to each constituent label; pure means single species, possibly several molecules.

Species	Old positive	New positive	New pure	Differential F1
CHO	7	9	2	0.5882
CO	37	44	28	0.8537
CO2	3	5	5	0.6000
COCHO	1	1	1	0.0000
COH	7	10	10	0.8235
COOH	9	9	9	0.5714
H	25	30	17	0.7843
O	7	7	7	0.4000
OCCOH	5	5	5	0.0000
OCHO	8	8	8	0.6154
OH	6	12	9	0.8182

The Cu–Au-only differential model gives micro F1 0.7668 on 101 single-species configurations and 0.5862 on 16 mixtures, with one exact mixture set. The joint model gives 0.7979 and 0.6333 on those same strata, with two exact mixture sets. The one clean negative is incorrectly assigned COOH and OCCOH by the Cu–Au-only differential model but correctly assigned the empty set by

the joint model. This is a scoped null-control result, not an estimated experimental false-positive rate. A label with one positive has no positive training example in the fold that holds it out.

The target for loading is CO per top-surface metal atom. Cu-normalized loading and CO/nm^2 remain separate deterministic conversions; they are not inputs to the regressor. Only CO-only configurations train the coverage models. Results on CO-containing mixtures assess interference, not independently validated general mixture inference. Unknown experimental surface fraction and gain can confound absolute loading even when identity-related spectral shape is retained.

Table S12: Joint CO-only loading results separated by substrate domain. Four Cu(100) cases cannot establish facet transfer; pooled correlation can reflect different domain loading ranges.

Input	Domain	N	MAE	R^2
Native normalized	CuAu	28	0.01093	0.9708
Native normalized	Cu100	4	0.04405	0.9592
Corrected raw	CuAu	28	0.01361	0.9549
Corrected raw	Cu100	4	0.05418	0.9318
$\Delta\mu$	CuAu	28	0.01588	0.9390
$\Delta\mu$	Cu100	4	0.04123	0.9610

S20 Paired training-set enlargement and threshold sensitivity

The control retains the original 99 Cu–Au outer-fold assignments and labels, adding 19 geometry-grouped configurations to the training pool. Original inner-training assignments remain paired, while additions receive deterministic geometry-hash folds. Both cohorts use the same current arrays and dependencies. All 198 original top-two source files are byte-identical; interpolation-grid roundoff relative to the historical model matrices is documented separately. The fixed recipe uses population-weighted top-two normalized-raw Combined features and 64 independent balanced ExtraTrees, leaf size one and square-root feature sampling, with three predeclared seeds. Depth, features and seed are not selected by held-out scores. These tests reuse established configurations and are not prospective confirmation.

Transferred thresholds are learned from the original training partitions; retuned thresholds are learned inside the enlarged training partitions. Fixed 0.5 results are also retained in the source study. Paired comparisons always use the same 99 queries, rather than comparing the full enlarged test population with the old one. Configuration bootstrap intervals condition on saved fitted predictions and omit refit, new-parent and experimental uncertainty.

Table S13: Fixed-recipe raw-XAS predictions on the same 99 configurations. No automatic depth selector is used.

Seed	Training / threshold	Micro F1	Macro F1	Exact set
20261020	99 training	0.7500	0.5711	0.6263
20261020	118 / retuned	0.7570	0.5867	0.6263
20261020	118 / transferred	0.7638	0.5811	0.6364
20261021	99 training	0.7246	0.5572	0.5960
20261021	118 / retuned	0.7512	0.5962	0.6162
20261021	118 / transferred	0.7642	0.6043	0.6566
20261022	99 training	0.7556	0.6103	0.5758
20261022	118 / retuned	0.7477	0.5608	0.6263
20261022	118 / transferred	0.7544	0.6011	0.6162

Mean micro F1 changes from 0.7434 to 0.7519 with inner retuning, and to 0.7608 with transferred thresholds. Mean macro F1 changes from 0.5795 to 0.5812 with retuning; mean exact-set accuracy

changes from 0.5993 to 0.6229. Each seed's paired retuned micro-F1 interval crosses zero. The same-25 CO-only comparison reduces MAE from 0.013971 to 0.011444, with a conditional difference interval of $[-0.006057, +0.000344]$. Both are encouraging point changes without established improvement. The 28-case MAE has a different query set. A corresponding frozen-recipe differential learning curve remains unperformed.

S21 Chemical interpretation and reproducible figure sources

The retained OCCO/2H_2O spectral case is assigned 2CO plus two water molecules from its actual calculation geometry. It is an unsupported-label control, not a positive bonded-OCCO training example. Earlier unfinished checkpoints are historical snapshots. No retained matched OCCO/2CO spectral pair is available to isolate C–C-bond formation.

The accompanying code recomputes 25 saved prediction files and all 28 reference metric triplets, verifies array/manifest/fold correspondence and signed subtraction, and rebuilds Figures 2–5 without retraining. Current prediction arrays, compact manifests, source audit tables and model protocols are pinned to repository commit 798e2c81b7a9afa684dc2a5625607ec866430163 in the package provenance. The repository reports 153 focused tests and 110 saved-model replays at that commit; the manuscript editing pass independently checks stored predictions rather than repeating the full repository suite or loading all training artifacts. Historical studies remain separately identified.

References

[1] W. Ni, H. Chen, N. Tang *et al.*, High-purity ethylene production via indirect carbon dioxide electrochemical reduction, *Nat. Commun.*, 2024, **15**, 6078, DOI: 10.1038/s41467-024-50522-7.

[2] S. A. Francis, J. M. Velazquez, I. M. Ferrer, D. A. Torelli, D. Guevarra, M. T. McDowell, K. Sun, X. Zhou, F. H. Saadi, J. John, M. H. Richter, F. P. Hyler, K. M. Papadantonakis, B. S. Brunschwig and N. S. Lewis, Reduction of aqueous CO_2 to 1-propanol at MoS_2 electrodes, *Chem. Mater.*, 2018, **30**, 4902–4908, DOI: 10.1021/acs.chemmater.7b04428.

[3] C. Zhan, F. Dattila, C. Rettenmaier, A. Herzog, M. Herran, T. Wagner, F. Scholten, A. Bergmann, N. López and B. Roldan Cuenya, Key intermediates and Cu active sites for CO_2 electroreduction to ethylene and ethanol, *Nat. Energy*, 2024, **9**, 1485–1496, DOI: 10.1038/s41560-024-01633-4.

[4] J. H. Montoya, C. Shi, K. Chan and J. K. Nørskov, Theoretical insights into a CO dimerization mechanism in CO_2 electroreduction, *J. Phys. Chem. Lett.*, 2015, **6**, 2032–2037, DOI: 10.1021/acs.jpclett.5b00722.

[5] J. Timoshenko and B. Roldan Cuenya, In situ/operando electrocatalyst characterization by X-ray absorption spectroscopy, *Chem. Rev.*, 2021, **121**, 882–961, DOI: 10.1021/acs.chemrev.0c00396.

[6] S.-C. Lin, C.-C. Chang, S.-Y. Chiu, H.-T. Pai, T.-Y. Liao, C.-S. Hsu, W.-H. Chiang, M.-K. Tsai and H. M. Chen, Operando time-resolved X-ray absorption spectroscopy reveals the chemical nature enabling highly selective CO_2 reduction, *Nat. Commun.*, 2020, **11**, 3525, DOI: 10.1038/s41467-020-17231-3.

[7] C. G. Morales-Guio, E. R. Cave, S. A. Nitopi, J. T. Feaster, L. Wang, K. P. Kuhl, A. Jackson, N. C. Johnson, D. N. Abram, T. Hatsukade, C. Hahn and T. F. Jaramillo, Improved CO_2 reduction activity towards C_{2+} alcohols on a tandem gold on copper electrocatalyst, *Nat. Catal.*, 2018, **1**, 764–771, DOI: 10.1038/s41929-018-0139-9.

[8] J. J. Rehr, J. J. Kas, F. D. Vila, M. P. Prange and K. Jorissen, Parameter-free calculations of X-ray spectra with FEFF9, *Phys. Chem. Chem. Phys.*, 2010, **12**, 5503–5513, DOI: 10.1039/B926434E.

[9] Y. Joly, X-ray absorption near-edge structure calculations beyond the muffin-tin approximation, *Phys. Rev. B*, 2001, **63**, 125120, DOI: 10.1103/PhysRevB.63.125120.

[10] G. Kresse and J. Furthmüller, Efficient iterative schemes for ab initio total-energy calculations using a plane-wave basis set, *Phys. Rev. B*, 1996, **54**, 11169–11186, DOI: 10.1103/PhysRevB.54.11169.

[11] VASP Wiki, Supercell core-hole calculations, `https://vasp.at/wiki/index.php/SCH_calculations`, accessed 21 October 2026.

[12] J. Vinson, J. J. Rehr, J. J. Kas and E. L. Shirley, Bethe–Salpeter equation calculations of core excitation spectra, *Phys. Rev. B*, 2011, **83**, 115106, DOI: 10.1103/PhysRevB.83.115106.

[13] J. Timoshenko, A. Anspoks, A. Cintins, A. Kuzmin, J. Purans and A. I. Frenkel, Neural network approach for characterizing structural transformations by X-ray absorption fine structure spectroscopy, *Phys. Rev. Lett.*, 2018, **120**, 225502, DOI: 10.1103/PhysRevLett.120.225502.

[14] N. Marcella, Y. Liu, J. Timoshenko, E. Guan, M. Luneau, T. Shirman, A. M. Plonka, J. E. S. van der Hoeven, J. Aizenberg, C. M. Friend and A. I. Frenkel, Neural network assisted analysis of bimetallic nanocatalysts using X-ray absorption near edge structure spectroscopy, *Phys. Chem. Chem. Phys.*, 2020, **22**, 18902–18910, DOI: 10.1039/D0CP02098B.

[15] S. B. Torrisi, M. R. Carbone, B. A. Rohr, J. H. Montoya, Y. Ha, J. Yano, S. K. Suram and L. Hung, Random forest machine learning models for interpretable X-ray absorption near-edge structure spectrum-property relationships, *npj Comput. Mater.*, 2020, **6**, 109, DOI: 10.1038/s41524-020-00376-6.

[16] T. David, N. K. Nik Aznan, K. Garside and T. J. Penfold, Towards the automated extraction of structural information from X-ray absorption spectra, *Digital Discovery*, 2023, **2**, 1461–1470, DOI: 10.1039/D3DD00101F.

[17] Y. Chen, C. Chen, I. Hwang, M. J. Davis, W. Yang, C. Sun, G.-H. Lee, D. McReynolds, D. Allan, J. Marulanda Arias, S. P. Ong and M. K. Y. Chan, Robust machine learning inference from X-ray absorption near edge spectra through featurization, *Chem. Mater.*, 2024, **36**, 2304–2313, DOI: 10.1021/acs.chemmater.3c02584.

[18] H. Jia, Y. Chen, G.-H. Lee, J. Smith, M. Chi, W. Yang and M. K. Y. Chan, Revealing local structures through machine-learning-fused multimodal spectroscopy, *ACS Nano*, 2026, **20**, 4228–4240, DOI: 10.1021/acsnano.5c16942.

[19] A. M. Bran, S. Cox, O. Schilter, C. Baldassari, A. D. White and P. Schwaller, Augmenting large language models with chemistry tools, *Nat. Mach. Intell.*, 2024, **6**, 525–535, DOI: 10.1038/s42256-024-00832-8.

[20] R. B. Canty, B. A. Koscher, M. A. McDonald and K. F. Jensen, Integrating autonomy into automated research platforms, *Digital Discovery*, 2023, **2**, 1259–1268, DOI: 10.1039/D3DD00135K.

[21] A. Vriza *et al.*, Multi-agentic AI framework for end-to-end atomistic simulations, *Digital Discovery*, 2026, **5**, 440–452, DOI: 10.1039/D5DD00435G.

[22] NERSC, The Superfacility API, `https://docs.nersc.gov/services/sfapi/`, accessed 21 October 2026.

[23] Y. Joly, *FDMNES User's Guide*, November 2022, `https://data-analysis.synchrotron-soleil.fr/tmp/software/fdmnes/fdmnes_Linux/Doc/Manual_Eng.pdf`, accessed 21 October 2026.

[24] N. Ahmed, T. Natarajan and K. R. Rao, Discrete cosine transform, *IEEE Trans. Comput.*, 1974, **C-23**, 90–93, DOI: 10.1109/T-C.1974.223784.

[25] P. Geurts, D. Ernst and L. Wehenkel, Extremely randomized trees, *Mach. Learn.*, 2006, **63**, 3–42, DOI: 10.1007/s10994-006-6226-1.

[26] J. H. Montoya, A. A. Peterson and J. K. Nørskov, Insights into C–C coupling in CO_2 electroreduction on copper electrodes, *ChemCatChem*, 2013, **5**, 737–742, DOI: 10.1002/cctc.201200564.

[27] D. E. Ramaker and D. C. Koningsberger, The atomic AXAFS and Delta mu XANES techniques as applied to heterogeneous catalysis and electrocatalysis, *Phys. Chem. Chem. Phys.*, 2010, **12**, 5514–5534, DOI: 10.1039/B927120C.

[28] D. E. Ramaker, A. Korovina, V. Croze, J. Melke and C. Roth, Following ORR intermediates adsorbed on a Pt cathode catalyst during break-in of a PEM fuel cell by in operando X-ray absorption spectroscopy, *Phys. Chem. Chem. Phys.*, 2014, **16**, 13645–13653, DOI: 10.1039/C4CP00192C.

[29] Q. Lei, L. Huang, J. Yin *et al.*, Structural evolution and strain generation of derived-Cu catalysts during CO_2 electroreduction, *Nat. Commun.*, 2022, **13**, 4857, DOI: 10.1038/s41467-022-32601-9.

[30] T.-C. Weng, G. S. Waldo and J. E. Penner-Hahn, A method for normalization of X-ray absorption spectra, *J. Synchrotron Rad.*, 2005, **12**, 506–510, DOI: 10.1107/S0909049504034193.

[31] L. Bugarin, H. A. Suarez Orduz and P. Glatzel, Area normalization of HERFD-XANES spectra, *J. Synchrotron Rad.*, 2024, **31**, 1118–1125, DOI: 10.1107/S1600577524005307.